

Electric Currents and **Ohm's** Law

- 7 A fuse in an electric circuit is a wire that is designed to melt, and thereby open the circuit, if the current exceeds a predetermined value. Suppose that the material to be used in a fuse melts when the current density rises to 440 A/cm^2 . What diameter of cylindrical wire should be used to make a fuse that will limit the current to 0.50 A ?
- 8 A small but measurable current of $1.2 \times 10^{-10} \text{ A}$ exists in a copper wire whose diameter is 2.5 mm . The number of charge carriers per unit volume is $8.49 \times 10^{28} \text{ m}^{-3}$. Assuming the current is uniform, calculate the (a) current density and (b) electron drift speed.
- 9 The magnitude $J(r)$ of the current density in a certain cylindrical wire is given as a function of radial distance from the center of the wire's cross section as $J(r) = Br$, where r is in meters, J is in amperes per square meter, and $B = 2.00 \times 10^5 \text{ A/m}^3$. This function applies out to the wire's radius of 2.00 mm . How much current is contained within the width of a thin ring concentric with the wire if the ring has a radial width of $10.0 \mu\text{m}$ and is at a radial distance of 1.20 mm ?

30. **II** a. How long must a 0.60-mm-diameter aluminum wire be to have a 0.50 A current when connected to the terminals of a 1.5 V flashlight battery?
- b. What is the current if the wire is half this length?
31. **II** The terminals of a 0.70 V watch battery are connected by a 100-m-long gold wire with a diameter of 0.10 mm. What is the current in the wire?
32. **BIO II** The femoral artery is the large artery that carries blood to the leg. What is the resistance of a 20-cm-long column of blood in a 1.0-cm-diameter femoral artery? The conductivity of blood is $0.63 \, \Omega^{-1} \text{m}^{-1}$.
33. **I** Household wiring often uses 2.0-mm-diameter copper wires. The wires can get rather long as they snake through the walls from the fuse box to the farthest corners of your house. What is the potential difference across a 20-m-long, 2.0-mm-diameter copper wire carrying an 8.0 A current?

DC Circuits: **Single-Loop** Circuits

Module 27-1 Single-Loop Circuits

•1 **SSM WWW** In Fig. 27-25, the ideal batteries have emfs $\mathcal{E}_1 = 12 \text{ V}$ and $\mathcal{E}_2 = 6.0 \text{ V}$. What are (a) the current, the dissipation rate in (b) resistor 1 (4.0Ω) and (c) resistor 2 (8.0Ω), and the energy transfer rate in (d) battery 1 and (e) battery 2? Is energy being supplied or absorbed by (f) battery 1 and (g) battery 2?

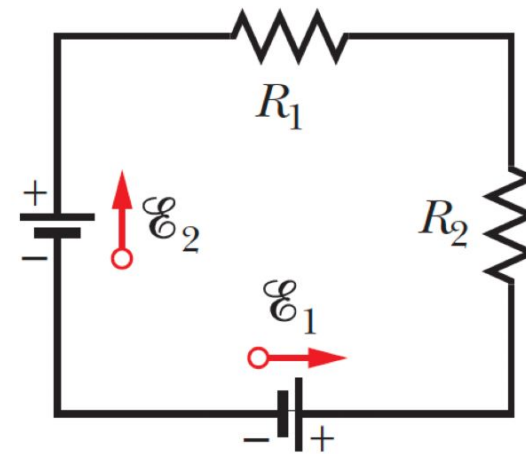


Figure 27-25
Problem 1.

•2 In Fig. 27-26, the ideal batteries have emfs $\mathcal{E}_1 = 150 \text{ V}$ and $\mathcal{E}_2 = 50 \text{ V}$ and the resistances are $R_1 = 3.0 \Omega$ and $R_2 = 2.0 \Omega$. If the potential at P is 100 V , what is it at Q ?

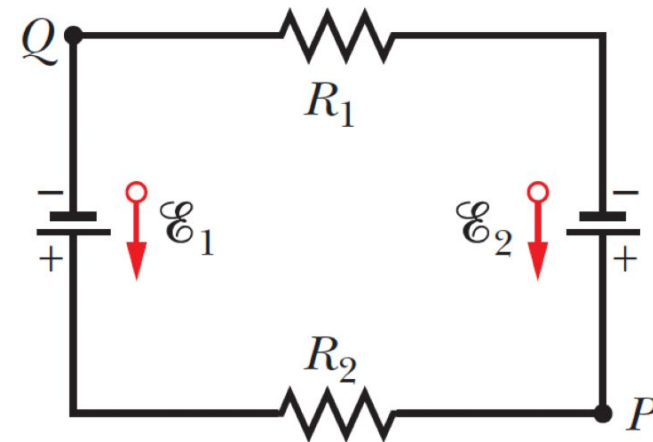


Figure 27-26 Problem 2.

•3 **ILW** A car battery with a 12 V emf and an internal resistance of 0.040Ω is being charged with a current of 50 A .

What are (a) the potential difference V across the terminals, (b) the rate P_r of energy dissipation inside the battery, and (c) the rate P_{emf} of energy conversion to chemical form? When the battery is used to supply 50 A to the starter motor, what are (d) V and (e) P_r ?

••15 **ILW** The current in a single-loop circuit with one resistance R is 5.0 A. When an additional resistance of $2.0\ \Omega$ is inserted in series with R , the current drops to 4.0 A. What is R ?

•••16 A solar cell generates a potential difference of 0.10 V when a $500\ \Omega$ resistor is connected across it, and a potential difference of 0.15 V when a $1000\ \Omega$ resistor is substituted. What are the (a) internal resistance and (b) emf of the solar cell? (c) The area of the cell is $5.0\ \text{cm}^2$, and the rate per unit area at which it receives energy from light is $2.0\ \text{mW}/\text{cm}^2$. What is the efficiency of the cell for converting light energy to thermal energy in the $1000\ \Omega$ external resistor?

•••17 **SSM** In Fig. 27-33, battery 1 has emf $\mathcal{E}_1 = 12.0\ \text{V}$ and internal resistance $r_1 = 0.016\ \Omega$ and battery 2 has emf $\mathcal{E}_2 = 12.0\ \text{V}$ and internal resistance $r_2 = 0.012\ \Omega$. The batteries are connected in series with an external resistance R . (a) What R value makes the terminal-to-terminal potential difference of one of the batteries zero? (b) Which battery is that?

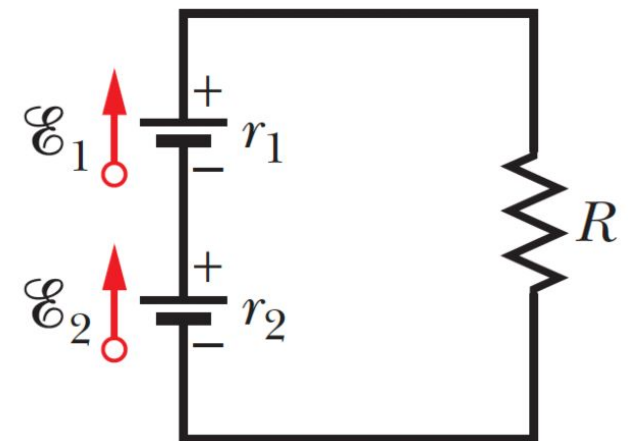


Figure 27-33
Problem 17.

DC Circuits: **Multi-Loop** Circuits

••30  In Fig. 27-41, the ideal batteries have emfs $\mathcal{E}_1 = 10.0 \text{ V}$ and $\mathcal{E}_2 = 0.500\mathcal{E}_1$, and the resistances are each $4.00 \text{ } \Omega$. What is the current in (a) resistance 2 and (b) resistance 3?

••31   In Fig. 27-42, the ideal batteries have emfs $\mathcal{E}_1 = 5.0 \text{ V}$ and $\mathcal{E}_2 = 12 \text{ V}$, the resistances are each $2.0 \text{ } \Omega$, and the potential is defined to be zero at the grounded point of the circuit. What are potentials (a) V_1 and (b) V_2 at the indicated points?

Figure 27-40 Problem 29.

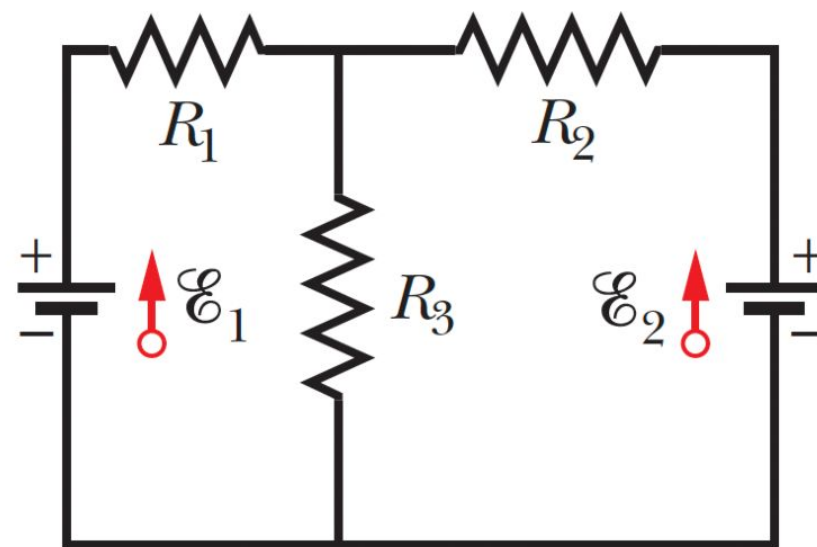


Figure 27-41 Problems 30, 41, and 88.

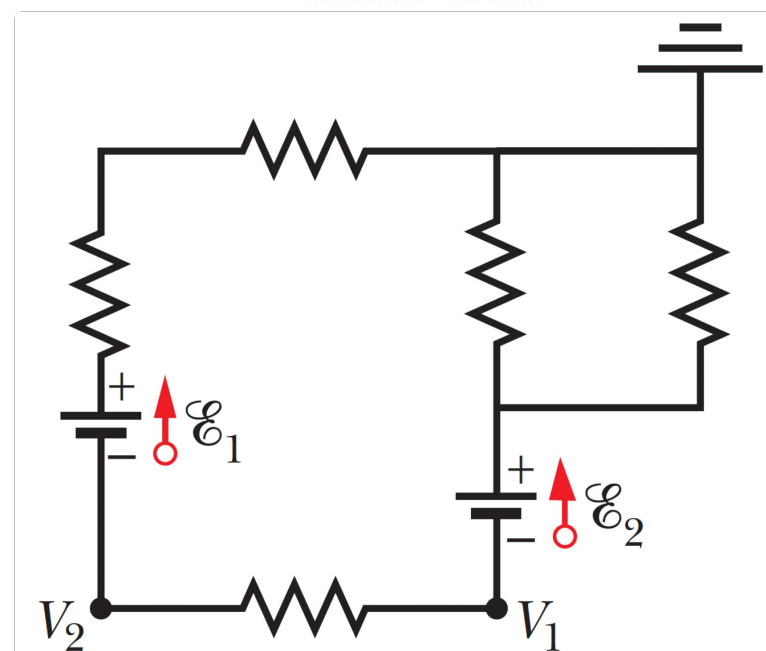


Figure 27-42 Problem 31.

- 69 • If the potential drop from point a to point b (Figure 25-52) is 12.0 V, find the current in each resistor. **SSM**
- 70 • If the potential drop between point a and point b (Figure 25-53) is 12.0 V, find the current in each resistor.

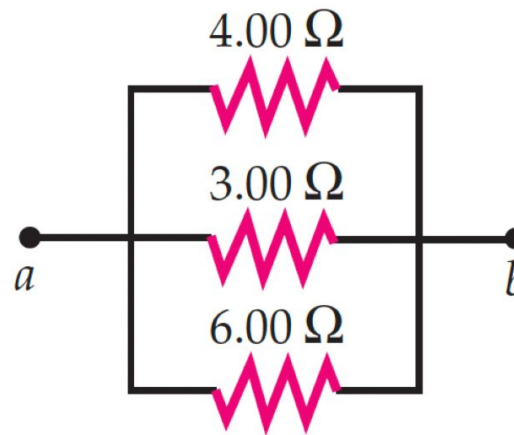


FIGURE 25-52 Problem 69

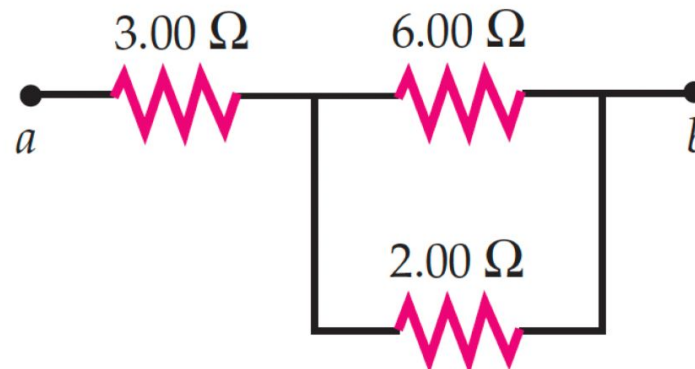


FIGURE 25-53 Problem 70

74 •• **ENGINEERING APPLICATION** You have been handed an unknown battery. Using your multimeter, you determine that when a $5.00\text{-}\Omega$ resistor is connected across the battery's terminals, the current in the battery is 0.500 A . When this resistor is replaced by an $11.0\text{-}\Omega$ resistor, the current drops to 0.250 A . From those data, find (a) the emf and (b) internal resistance of the battery.

75 •• (a) Find the equivalent resistance between point a and point b in Figure 25-56. (b) If the potential drop between point a and point b is 12.0 V , find the current in each resistor.

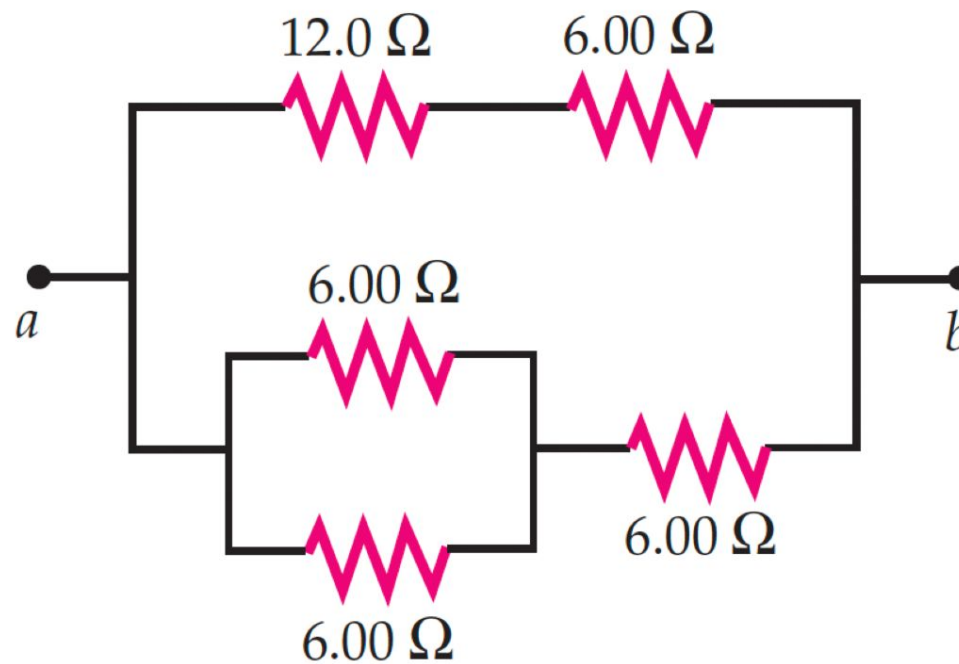


FIGURE 25-56 Problem 75

••33  In Fig. 27-44, the current in resistance 6 is $i_6 = 1.40\text{ A}$ and the resistances are $R_1 = R_2 = R_3 = 2.00\ \Omega$, $R_4 = 16.0\ \Omega$, $R_5 = 8.00\ \Omega$, and $R_6 = 4.00\ \Omega$. What is the emf of the ideal battery?

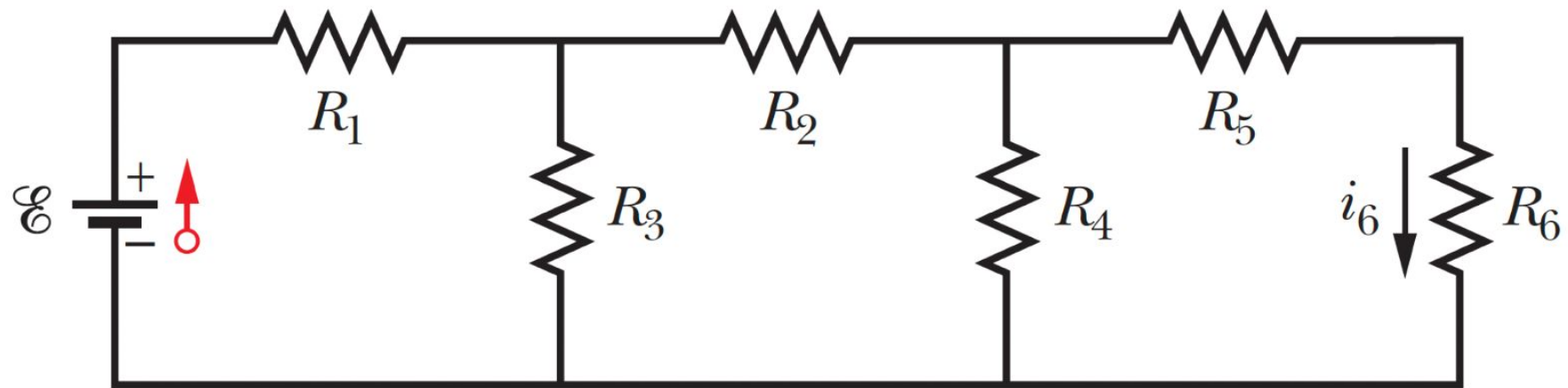


Figure 27-44 Problem 33.

••34 The resistances in Figs. 27-45*a* and *b* are all $6.0\ \Omega$, and the batteries are ideal $12\ \text{V}$ batteries. (a) When switch S in Fig. 27-45*a* is closed, what is the change in the electric potential V_1 across resistor 1, or does V_1 remain the same? (b) When switch S in Fig. 27-45*b* is closed, what is the change in V_1 across resistor 1, or does V_1 remain the same?

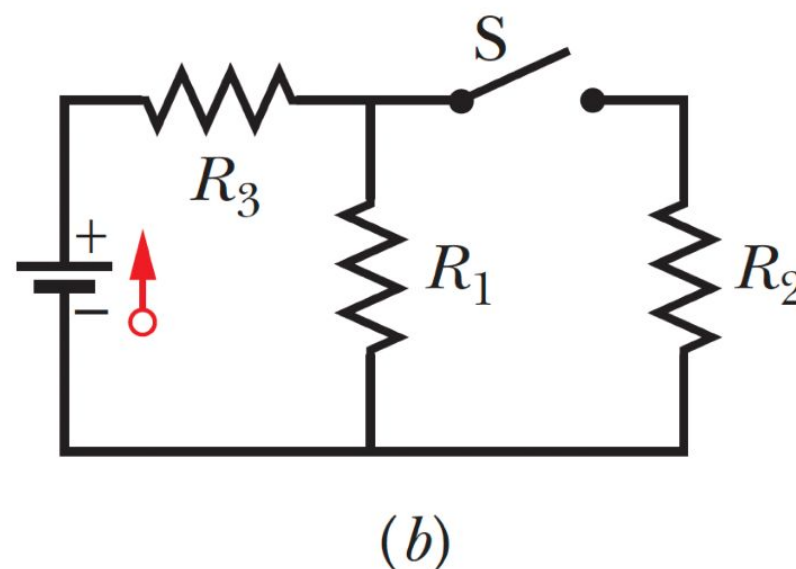
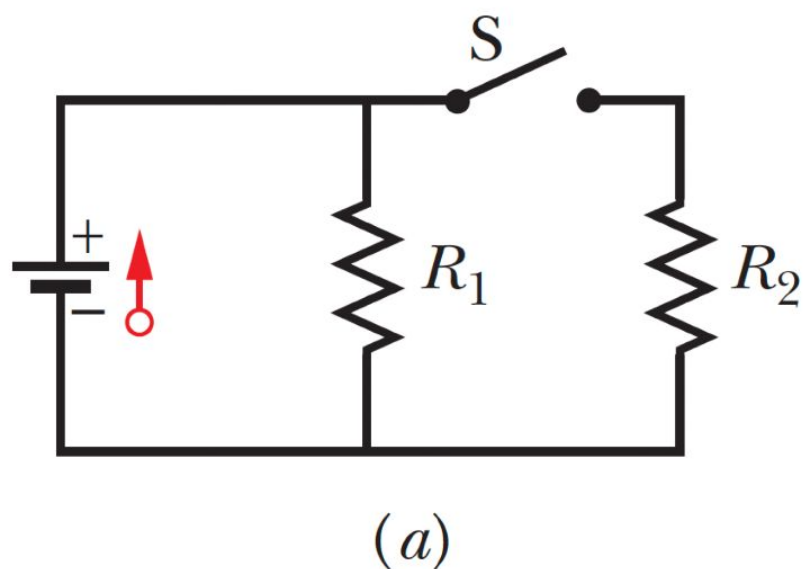
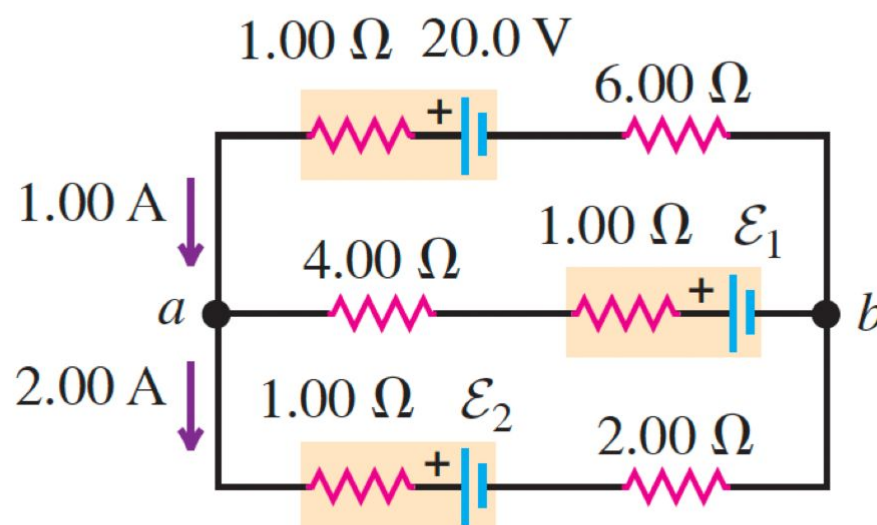


Figure 27-45 Problem 34.

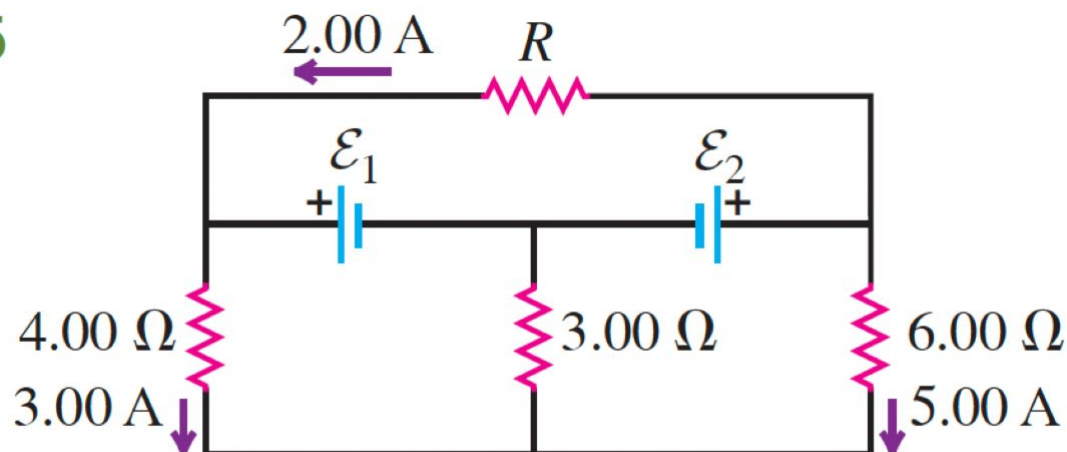
26.24 • Find the emfs $\mathcal{E}_1$ and $\mathcal{E}_2$ in the circuit of **Fig. E26.24**, and find the potential difference of point b relative to point a .

Figure **E26.24**



26.25 • In the circuit shown in **Fig. E26.25**, find (a) the current in the $3.00\ \Omega$ resistor; (b) the unknown emfs $\mathcal{E}_1$ and $\mathcal{E}_2$; (c) the resistance R . Note that three currents are given.

Figure **E26.25**



••45 **ILW** In Fig. 27-54, the resistances are $R_1 = 1.0 \, \Omega$ and $R_2 = 2.0 \, \Omega$, and the ideal batteries have emfs $\mathcal{E}_1 = 2.0 \, \text{V}$ and $\mathcal{E}_2 = \mathcal{E}_3 = 4.0 \, \text{V}$. What are the (a) size and (b) direction (up or down) of the current in battery 1, the (c) size and (d) direction of the current in battery 2, and the (e) size and (f) direction of the current in battery 3? (g) What is the potential difference $V_a - V_b$?

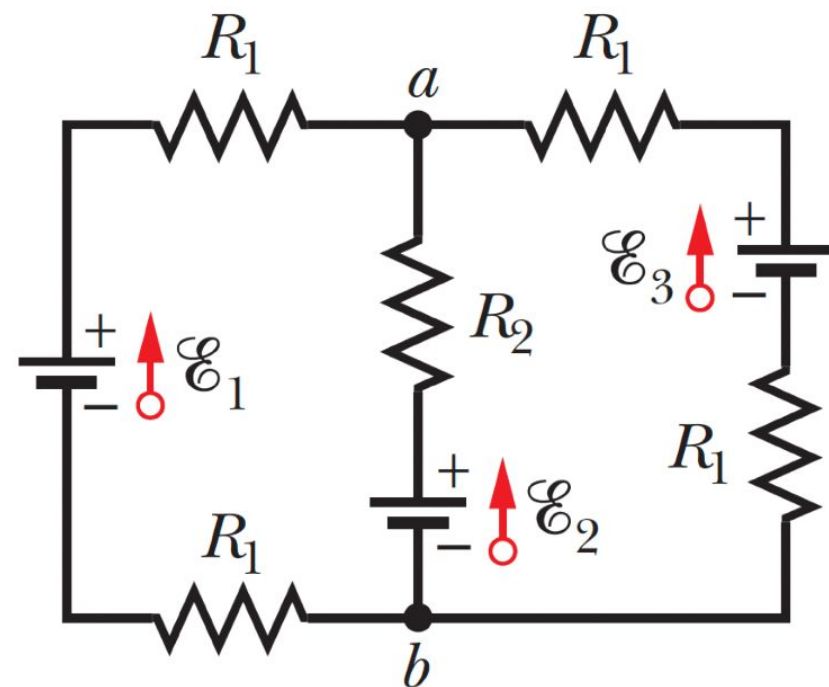


Figure 27-54 Problem 45.

Capacitors and Capacitance

Module 25-1 Capacitance

- 1 The two metal objects in Fig. 25-24 have net charges of $+70 \text{ pC}$ and -70 pC , which result in a 20 V potential difference between them. (a) What is the capacitance of the system? (b) If the charges are changed to $+200 \text{ pC}$ and -200 pC , what does the capacitance become? (c) What does the potential difference become?

Figure 25-24 Problem 1.



- 2 The capacitor in Fig. 25-25 has a capacitance of $25 \mu\text{F}$ and is initially uncharged. The battery provides a potential difference of 120 V . After switch S is closed, how much charge will pass through it?

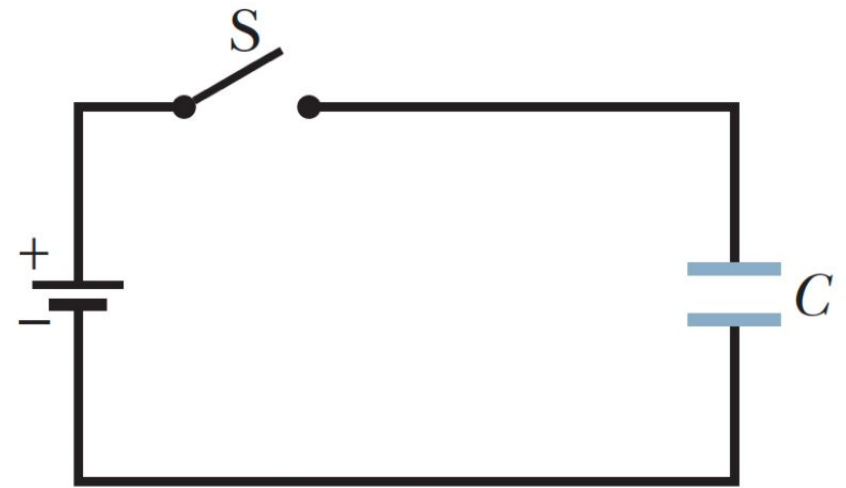


Figure 25-25 Problem 2.

24.23 • A $5.80\ \mu\text{F}$, parallel-plate, air capacitor has a plate separation of $5.00\ \text{mm}$ and is charged to a potential difference of $400\ \text{V}$. Calculate the energy density in the region between the plates, in units of J/m^3 .

24.24 • A parallel-plate air capacitor has a capacitance of $920\ \text{pF}$. The charge on each plate is $3.90\ \mu\text{C}$. (a) What is the potential difference between the plates? (b) If the charge is kept constant, what will be the potential difference if the plate separation is doubled? (c) How much work is required to double the separation?

24.1 • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of 4.00×10^6 V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

24.2 • The plates of a parallel-plate capacitor are 3.46 mm apart, and each has an area of 11.2 cm^2 . Each plate carries a charge of magnitude 6.80×10^{-8} C. The plates are in vacuum. What is (a) the capacitance; (b) the potential difference between the plates; (c) the magnitude of the electric field between the plates?

24.3 • A parallel-plate air capacitor of capacitance 246 pF has a charge of magnitude $0.149 \mu\text{C}$ on each plate. The plates are 0.371 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?

••14  In Fig. 25-30, the battery has a potential difference of $V = 10.0 \text{ V}$ and the five capacitors each have a capacitance of $10.0 \mu\text{F}$. What is the charge on (a) capacitor 1 and (b) capacitor 2?

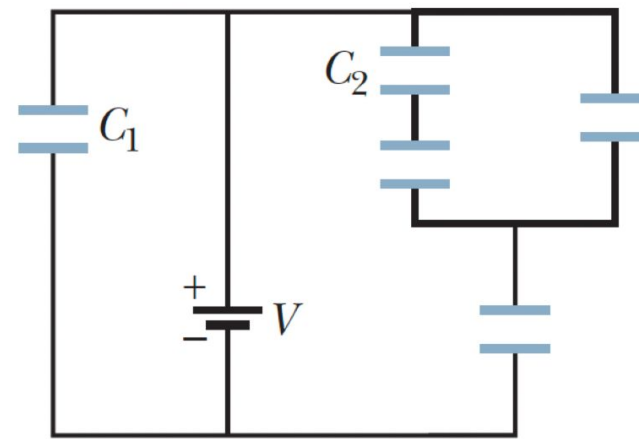


Figure 25-30 Problem 14.

••15  In Fig. 25-31, a 20.0 V battery is connected across capacitors of capacitances $C_1 = C_6 = 3.00 \mu\text{F}$ and $C_3 = C_5 = 2.00C_2 = 2.00C_4 = 4.00 \mu\text{F}$. What are (a) the equivalent capacitance C_{eq} of the capacitors and (b) the charge stored by C_{eq} ? What are (c) V_1 and (d) q_1 of capacitor 1, (e) V_2 and (f) q_2 of capacitor 2, and (g) V_3 and (h) q_3 of capacitor 3?

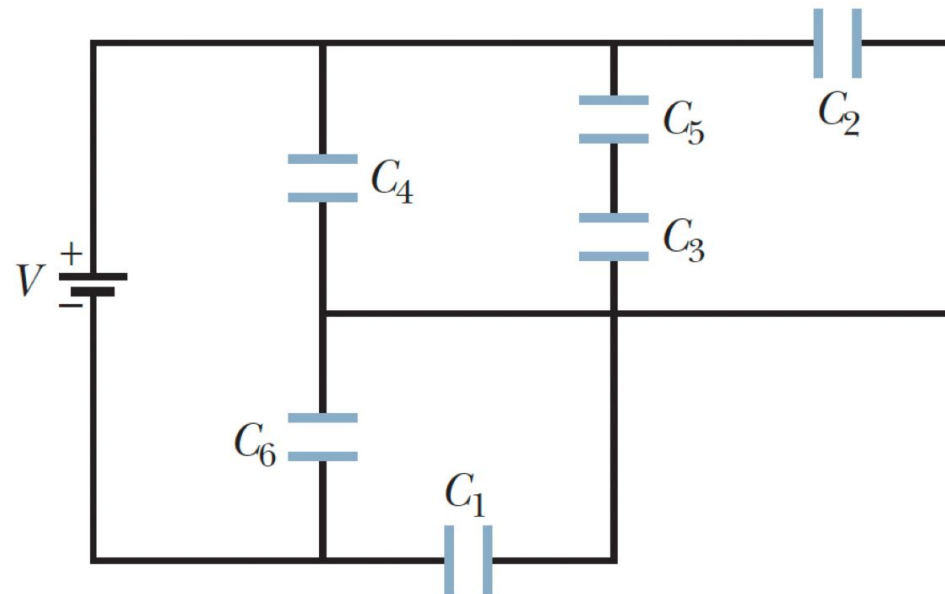


Figure 25-31 Problem 15.

- 11** **ILW** In Fig. 25-29, find the equivalent capacitance of the combination. Assume that $C_1 = 10.0 \mu\text{F}$, $C_2 = 5.00 \mu\text{F}$, and $C_3 = 4.00 \mu\text{F}$.

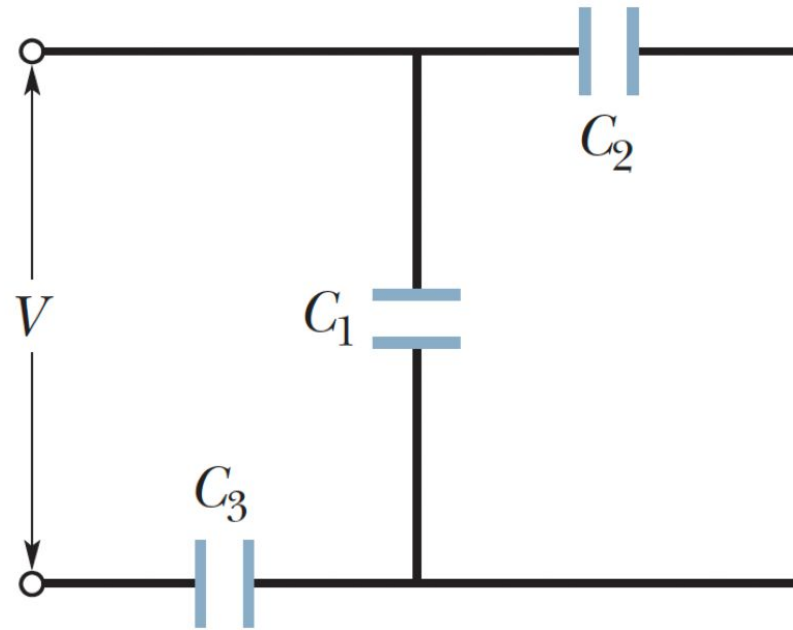


Figure 25-29 Problems 11, 17, and 38.

- 12** Two parallel-plate capacitors, $6.0 \mu\text{F}$ each, are connected in parallel to a 10 V battery. One of the capacitors is then squeezed so that its plate separation is 50.0% of its initial value. Because of the squeezing, (a) how much additional charge is transferred to the capacitors by the battery and (b) what is the increase in the total charge stored on the capacitors?

••19  In Fig. 25-34, the battery has potential difference $V = 9.0$ V, $C_2 = 3.0 \mu\text{F}$, $C_4 = 4.0 \mu\text{F}$, and all the capacitors are initially uncharged. When switch S is closed, a total charge of $12 \mu\text{C}$ passes through point a and a total charge of $8.0 \mu\text{C}$ passes through point b . What are (a) C_1 and (b) C_3 ?

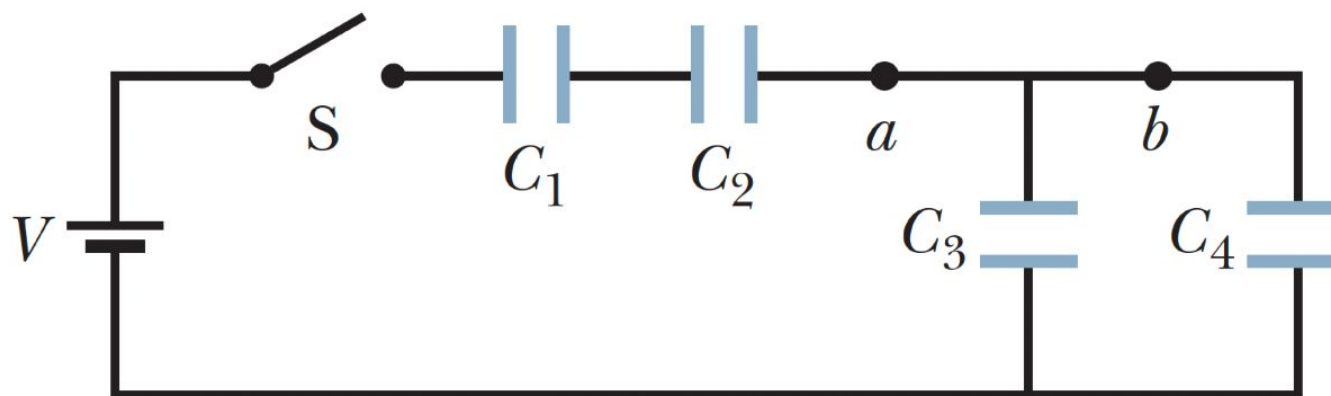


Figure 25-34 Problem 19.

Energy storage in Capacitors

- 29** What capacitance is required to store an energy of $10 \text{ kW} \cdot \text{h}$ at a potential difference of 1000 V ?
- 30** How much energy is stored in 1.00 m^3 of air due to the “fair weather” electric field of magnitude 150 V/m ?
- 31** **SSM** A $2.0 \mu\text{F}$ capacitor and a $4.0 \mu\text{F}$ capacitor are connected in parallel across a 300 V potential difference. Calculate the total energy stored in the capacitors.
- 32** A parallel-plate air-filled capacitor having area 40 cm^2 and plate spacing 1.0 mm is charged to a potential difference of 600 V . Find (a) the capacitance, (b) the magnitude of the charge on each plate, (c) the stored energy, (d) the electric field between the plates, and (e) the energy density between the plates.

••37 **SSM** **ILW** **WWW** The parallel plates in a capacitor, with a plate area of 8.50 cm^2 and an air-filled separation of 3.00 mm , are charged by a 6.00 V battery. They are then disconnected from the battery and pulled apart (without discharge) to a separation of 8.00 mm . Neglecting fringing, find (a) the potential difference between the plates, (b) the initial stored energy, (c) the final stored energy, and (d) the work required to separate the plates.

31. **I** To what potential should you charge a $1.0\ \mu\text{F}$ capacitor to store $1.0\ \text{J}$ of energy?
32. **II** A 2.0-cm -diameter parallel-plate capacitor with a spacing of $0.50\ \text{mm}$ is charged to $200\ \text{V}$. What are (a) the total energy stored in the electric field and (b) the energy density?
33. **II** $100\ \text{pJ}$ of energy is stored in a $1\ \text{cm} \times 1\ \text{cm} \times 1\ \text{cm}$ region of uniform electric field. What is the electric field strength?
34. **I** The $90\ \mu\text{F}$ capacitor in a defibrillator unit supplies an average of $6500\ \text{W}$ of power to the chest of the patient during a discharge lasting $5.0\ \text{ms}$. To what voltage is the capacitor charged?

BIO
INT

••45 A certain parallel-plate capacitor is filled with a dielectric for which $\kappa = 5.5$. The area of each plate is 0.034 m^2 , and the plates are separated by 2.0 mm . The capacitor will fail (short out and burn up) if the electric field between the plates exceeds 200 kN/C . What is the maximum energy that can be stored in the capacitor?

••46 In Fig. 25-46, how much charge is stored on the parallel-plate capacitors by the 12.0 V battery? One is filled with air, and the other is filled with a dielectric for which $\kappa = 3.00$; both capacitors have a plate area of $5.00 \times 10^{-3} \text{ m}^2$ and a plate separation of 2.00 mm .

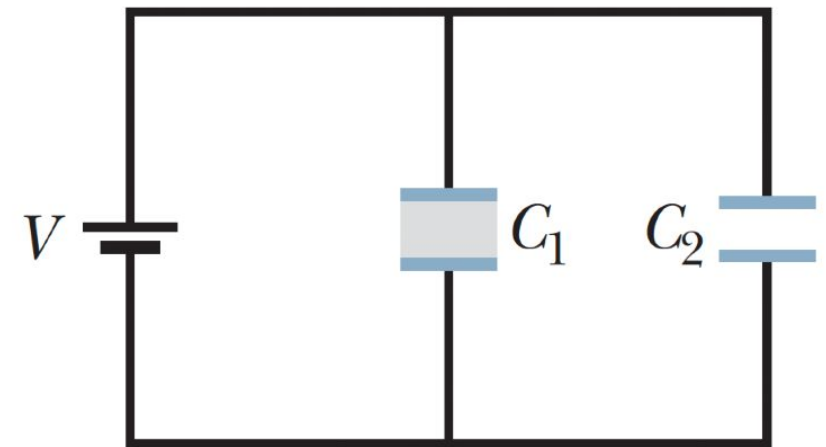


Figure 25-46 Problem 46.

73 Figure 25-58 shows a four-capacitor arrangement that is connected to a larger circuit at points A and B . The capacitances are $C_1 = 10\ \mu\text{F}$ and $C_2 = C_3 = C_4 = 20\ \mu\text{F}$. The charge on capacitor 1 is $30\ \mu\text{C}$. What is the magnitude of the potential difference $V_A - V_B$?

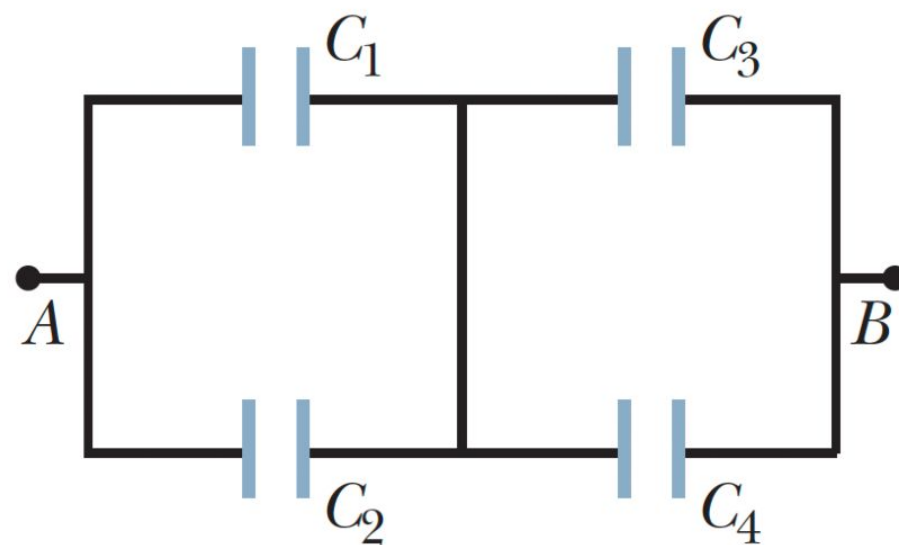


Figure 25-58 Problem 73.

74 You have two plates of copper, a sheet of mica (thickness = $0.10\ \text{mm}$, $\kappa = 5.4$), a sheet of glass (thickness = $2.0\ \text{mm}$, $\kappa = 7.0$), and a slab of paraffin (thickness = $1.0\ \text{cm}$, $\kappa = 2.0$). To make a parallel-plate capacitor with the largest C , which sheet should you place between the copper plates?

75 A capacitor of unknown capacitance C is charged to $100\ \text{V}$ and connected across an initially uncharged $60\ \mu\text{F}$ capacitor. If the final potential difference across the $60\ \mu\text{F}$ capacitor is $40\ \text{V}$, what is C ?

Transient DC Circuits: Series RC Circuits

•**57** Switch S in Fig. 27-63 is closed at time $t = 0$, to begin charging an initially uncharged capacitor of capacitance $C = 15.0 \mu\text{F}$ through a resistor of resistance $R = 20.0 \Omega$. At what time is the potential across the capacitor equal to that across the resistor?

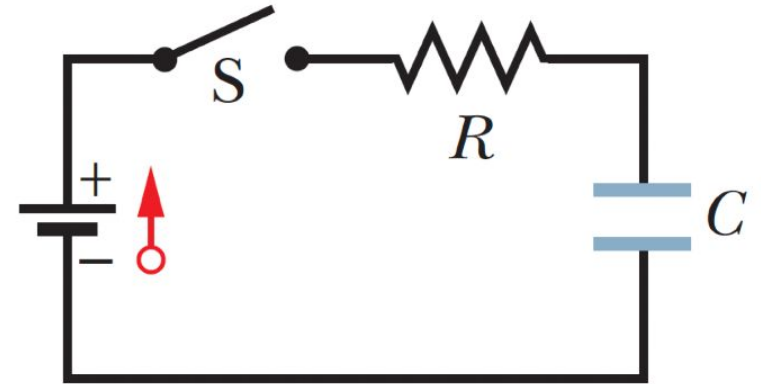


Figure 27-63 Problems 57 and 96.

•**58** In an RC series circuit, emf $\mathcal{E} = 12.0 \text{ V}$, resistance $R = 1.40 \text{ M}\Omega$, and capacitance $C = 1.80 \mu\text{F}$. (a) Calculate the time constant. (b) Find the maximum charge that will appear on the capacitor during charging. (c) How long does it take for the charge to build up to $16.0 \mu\text{C}$?

- 59** **SSM** What multiple of the time constant τ gives the time taken by an initially uncharged capacitor in an RC series circuit to be charged to 99.0% of its final charge?
- 60** A capacitor with initial charge q_0 is discharged through a resistor. What multiple of the time constant τ gives the time the capacitor takes to lose (a) the first one-third of its charge and (b) two-thirds of its charge?
- 61** **ILW** A $15.0\text{ k}\Omega$ resistor and a capacitor are connected in series, and then a 12.0 V potential difference is suddenly applied across them. The potential difference across the capacitor rises to 5.00 V in $1.30\text{ }\mu\text{s}$. (a) Calculate the time constant of the circuit. (b) Find the capacitance of the capacitor.